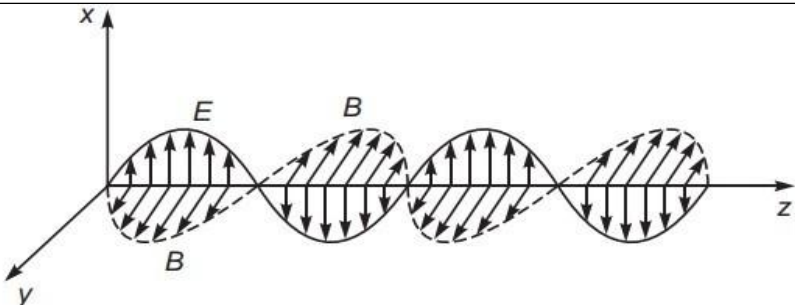


KENDRIYA VIDYALAYA SANGATHAN VARANASI REGION
PRE-BOARD-I EXAMINATION: 2024 – 25
MARKING SCHEME
CLASS-XII (PHYSICS)

Q.NO	ANSWER	Marks
1	(b) F	1
2	(b) $2 \times 10^{-8} \Omega \text{ m}$	1
3	(c) For a uniform electric field, they are concentric spheres.	1
4	(b) repel to each other	1
5	(a) 220 V, 2.2 A	1
6	(b) 10^{-6} T	1
7	(b) Coherent	1
8	(c) intensity of image will decrease but complete image is formed	1
9	(c) 6 cm	1
10	a) its wavelength decreases	1
11	(c) 3000 A^0	1
12	(c) lowers the potential barrier	1
13	(c) If Assertion is true but Reason is false.	1
14	b) If both Assertion and Reason are true but Reason is not the correct explanation of Assertion..	1
15	a) If both Assertion and Reason are true and Reason is correct explanation of Assertion	1
16	a) If both Assertion and Reason are true and Reason is correct explanation of Assertion	1
17	Workdone $W = 2pE$ Calculation $W = 2.4 \times 10^{-6} \text{ J}$	1 $\frac{1}{2}$ $\frac{1}{2}$
18	 $E = E_0 \cdot \sin(\omega t - kz)$ and $B = B_0 \cdot \sin(\omega t - kz)$ OR Radio wave, X rays, γ rays , visible light	1 $\frac{1}{2} + \frac{1}{2}$ ($\frac{1}{2} + \frac{1}{2}$ $+ \frac{1}{2} + \frac{1}{2}$)
19	Emf, $e = -N (\Phi_2 - \Phi_1)/t$ Calculation $E = -200 \text{ V}$	1 $\frac{1}{2}$ $\frac{1}{2}$
20	Correct Definition Metal A has greater threshold wavelength Justification $E \propto 1/\lambda$	1 $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$
21	Correct Graph Correct graph	1 1
22	Correct statements Balance condition of wheat stone bridge- $P/Q = R/S$ $R = 200 \text{ Ohm}$	2 $\frac{1}{2}$ $\frac{1}{2}$

23	The average velocity with which free electrons drifted towards the positive end of the wire in the presence of external electric field and opposite to electric field. Derivation $I = \frac{Q}{t} = \frac{Aen}{l} = neAv_d$ Current density, $j = \frac{I}{A} = nev_d$ Where, n and e are constants $\therefore j \propto v_d$ i.e., current density is directly proportional to the drift velocity	1 1 1/2 1/2
24	Phasor Diagram Correct derivation Correct graph	1/2 2 1/2
25	Ray diagram For Correct derivation $\mu = \frac{\sin\left(\frac{A+\delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$ Graph between i and δ	1/2 2 1/2
26	Focal length of the objective lens, $f_1 = 2.0$ cm Focal length of the eyepiece, $f_2 = 6.25$ cm Distance between the objective lens and the eyepiece, $d = 15$ cm Least distance of distinct vision, $d' = 25$ cm $\therefore$ Image distance for the eyepiece, $v_2 = -25$ cm Object distance for the eyepiece = u_2 Using lens formula, we have the relation: $u_2 = -5$ cm Image distance for the objective lens, $v_1 = 10$ cm Object distance for the objective lens = u_1 According to the lens formula, we have the relation: Magnitude of the object distance, $u_1 = 2.5$ cm Using formula of magnifying Hence, the magnifying power of the microscope is 20.	1 1 1 1

	OR (i) Correct diagram in each case (ii) $1/f = 1/f_1 + 1/f_2$ Calculation for $f = -12\text{ cm}$	1 1
27	Correct circuit diagram Principle of rectifier Correct wave forms	1 1 $\frac{1}{2} + \frac{1}{2}$
28	(i) Correct definition (ii) Bulb B1 will glow as diode D1 is forward biased so current is passed through B1 Bulb B2 will not glow as D2 is reverse biased so current will not flow through B2	1 1 1
29	(i) c (ii) b (iii) d (iv) a OR c	1 1 1 1
30	(i) b (ii) a (iii) d (iv) a OR b	1 1 1 1
31	(i) Derivation for electric field inside the sphere with suitable diagram. (ii) By $\sigma = q / 4\pi R^2$ Calculation for $q = 7.85 \times 10^{-3}\text{ C}$ $\Phi = q / \epsilon_0$ Calculation for $\Phi = 8.87 \times 10^8\text{ Nm}^2/\text{C}$ OR (i) Diagram for axial position Derivation for electric field in axial position.. Electric field at midpoint is zero. (ii) $\Phi = q / \epsilon_0$ Calculation for $\Phi_1 / \Phi_2 = 1/3$ No change as the medium is outside S_1	2 $\frac{1}{2}$ 1 $\frac{1}{2}$ 1 $\frac{1}{2}$ 2 $\frac{1}{2}$ $\frac{1}{2}$ 1 $\frac{1}{2}$

32	(i) Statement of Amperes law Derivation of magnetic field with diagram (ii) Correct Graph with labelling of axes. (iii) Calculation for $B = 1.2 \times 10^{-6} \text{ T}$ Direction - Vertically upward OR (i) Derivation for force between two parallel conductors with diagram. Definition of Ampere (ii) Calculation for $F_1 = 8 \times 10^{-7} \text{ N}$ (Attractive) $F_2 = 2.6 \times 10^{-7} \text{ N}$ (Repulsive) Net force $F = F_1 - F_2 = 5.4 \times 10^{-7} \text{ N}$ (Attractive)	$\frac{1}{2}$ 1 $\frac{1}{2}$ 1 1 $\frac{1}{2}$ $\frac{1}{2}$ 2 1 2
33	(i) Correct derivation of Snell's law with diagram. Correct diagrams for (a) and (b) (ii) Identification of $i = 45^\circ$ $\mu = 1/\sin i$ Calculation of $\mu = 1.414$ OR (i) Ray diagram of compound microscope Expression of magnifying power (ii) For path diff = λ, phase difference = 2π By using $I' = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \phi$ For $I_1 = I_2 = I$ Resultant intensity = $4I = K$ (given) Now for path diff = $\lambda/3$, phase diff = $2\pi/3$ New intensity $I' = I = K/4$	2 $\frac{1}{2} + \frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ 1 1 $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ 1 $\frac{1}{2}$ $\frac{1}{2}$